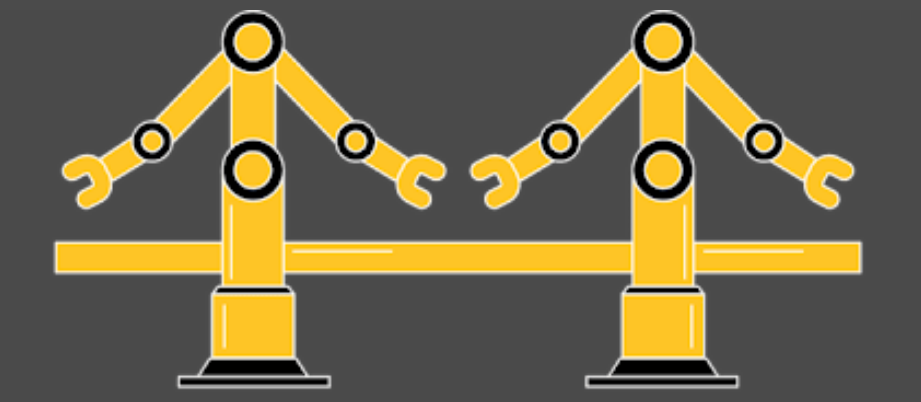


Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion.



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Background

Autonomous driving in cold regions faces significant challenges in snow-covered areas, where traditional navigation stacks degrade. Environment features, including lane lines, road boundaries, and road structures are often obscured by snow, transforming the task into off-road navigation.

Observation

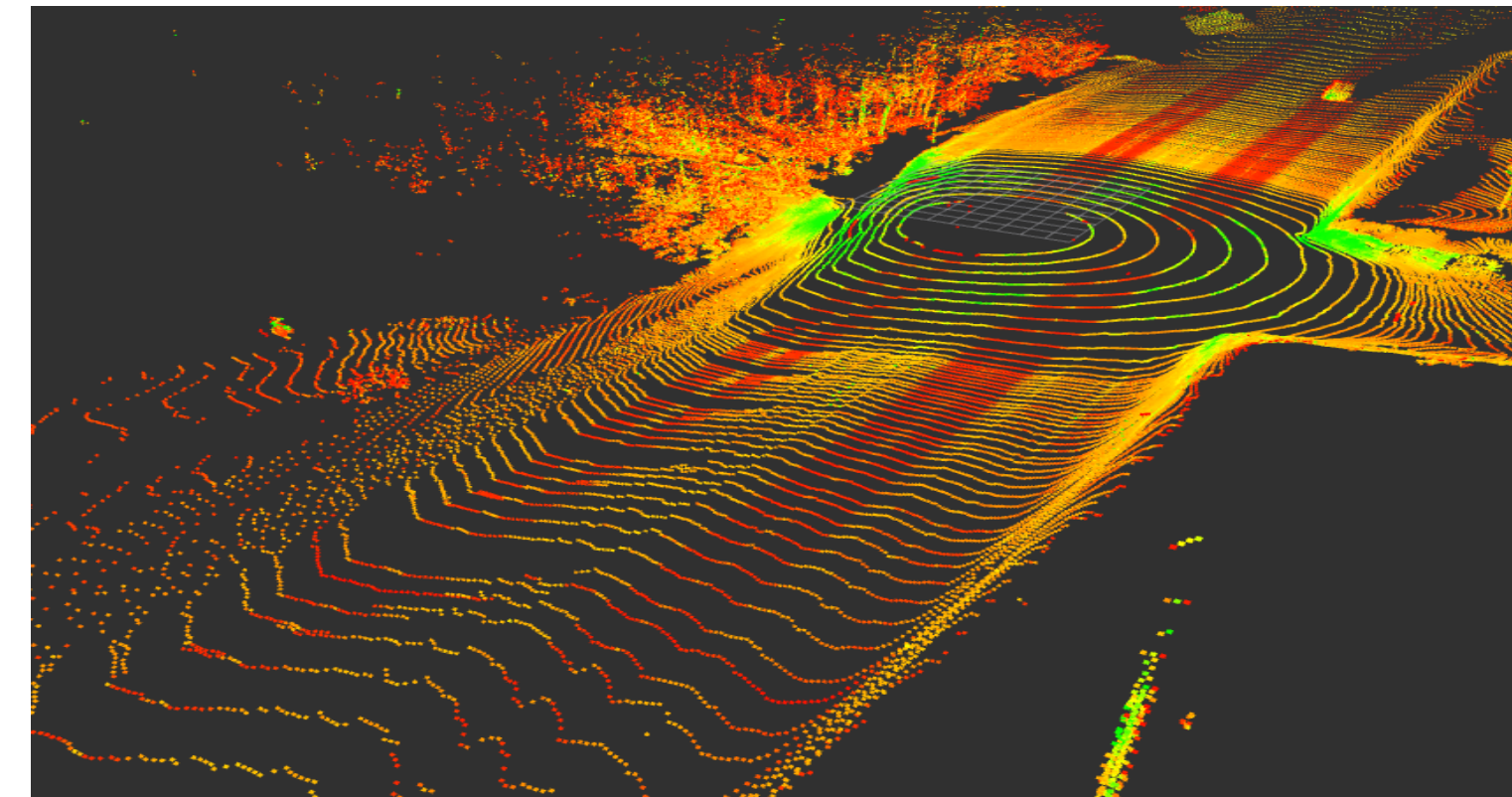
In such conditions, human drivers usually follow existing wheel tracks because wheel tracks:

1. Were verified by previous vehicles (traversability checked)
2. Provide more friction (more stability)
3. Provide extra lateral force (more stability)

Existing Research



Image-based



LiDAR-based

Challenges & Solution



RGB image

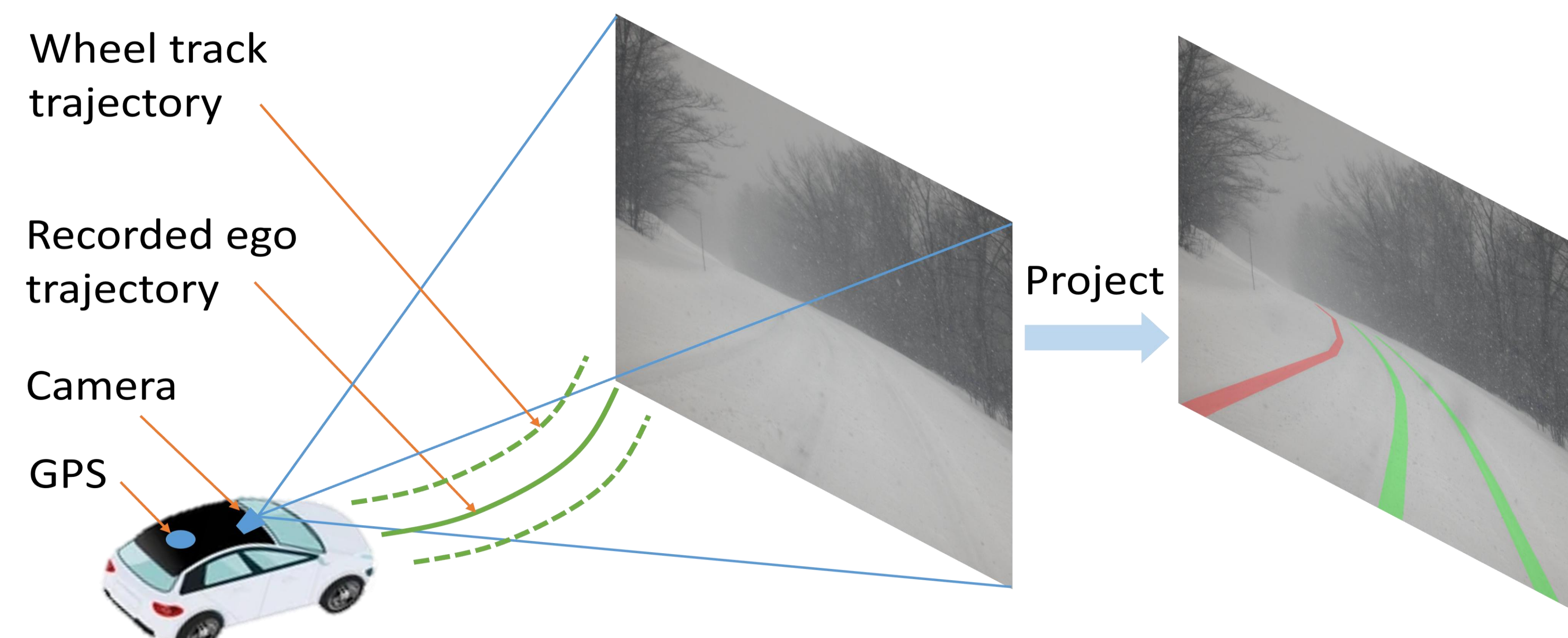
Featureless:

- Whiteout
- Low contrast
- Wheel track degradation

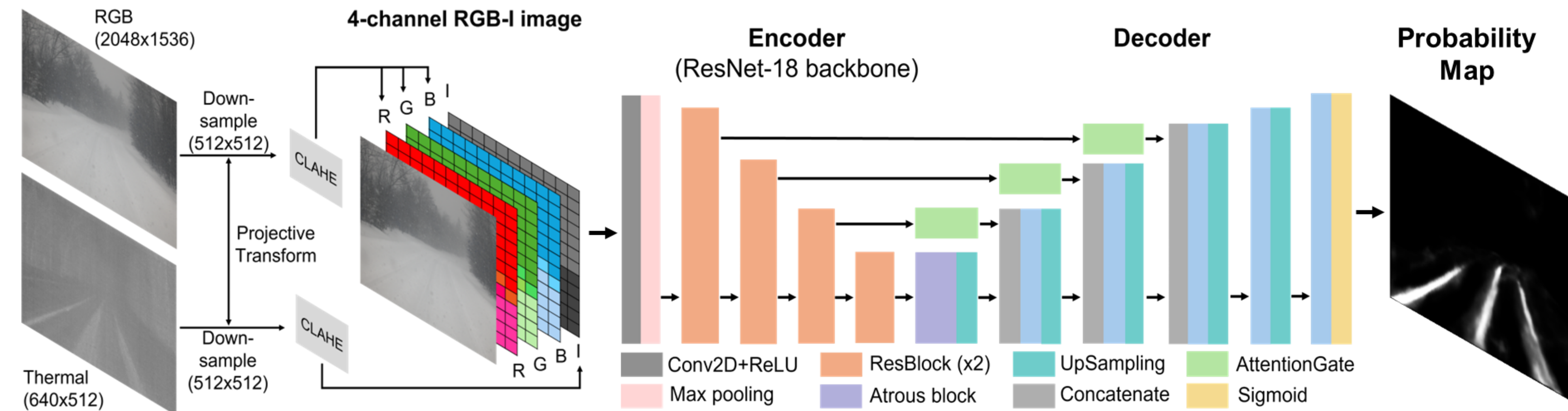


Thermal image

Self-supervised Labeling



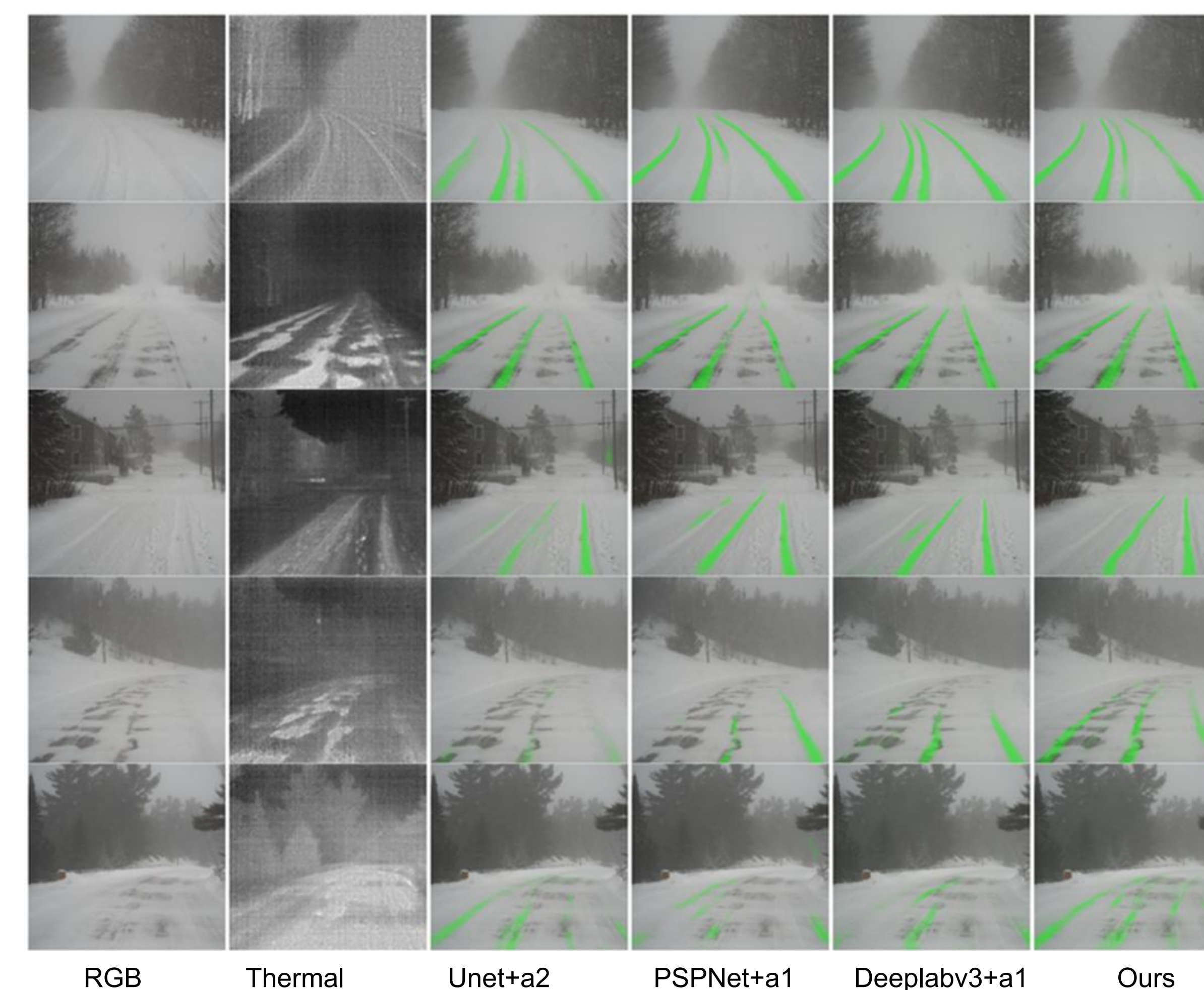
Proposed Method



Results

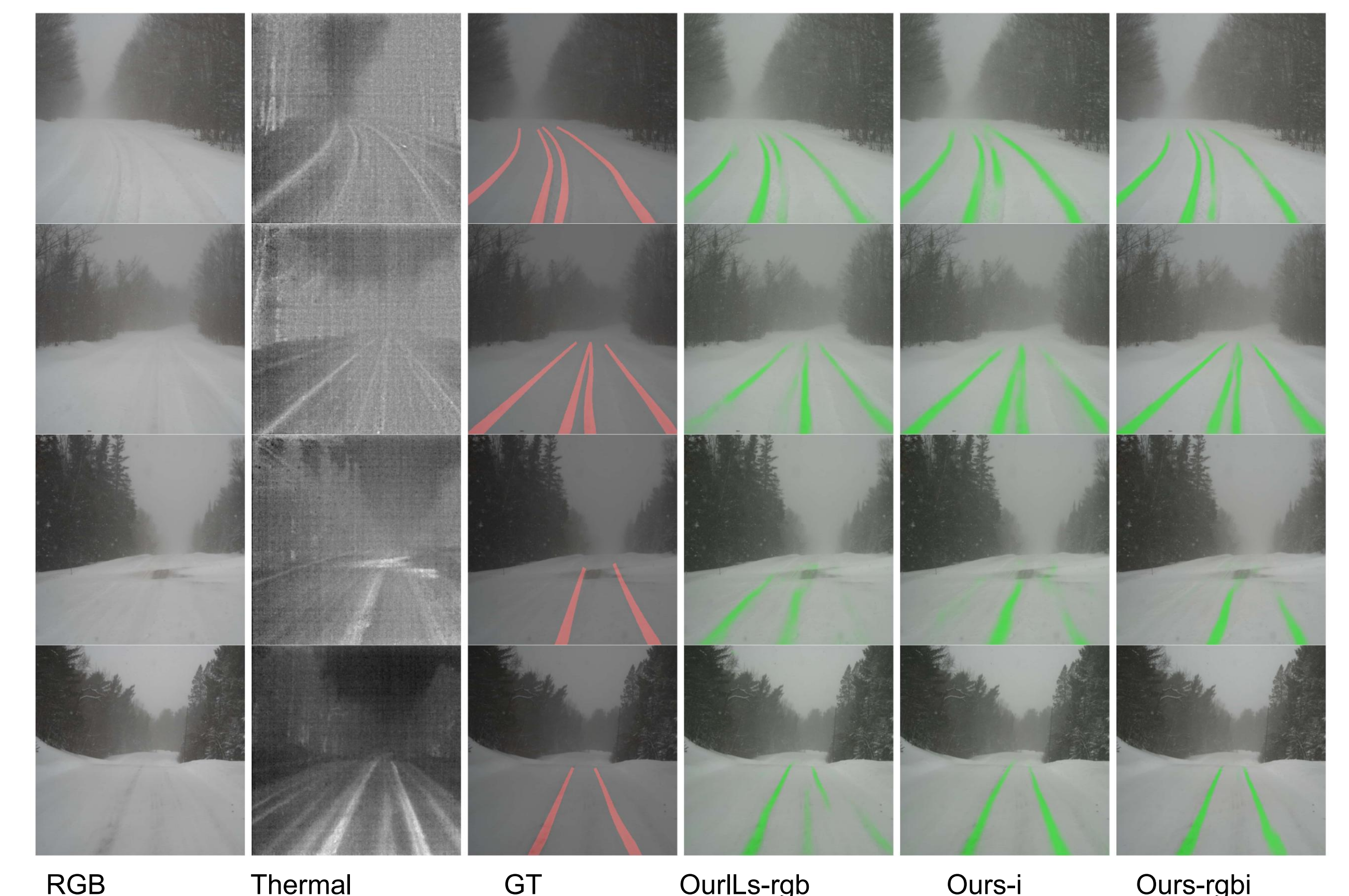
Overall Comparison

Model	AUROC	MaxF	AP	PRE	REC	FPR	FNR
UNet	0.7068	0.5197	0.6079	0.7691	0.4192	0.0056	0.5808
PSPNet	0.7532	0.6102	0.6666	0.7978	0.5122	0.0057	0.4878
DeepLabv3+	0.7446	0.6002	0.6629	0.8074	0.4943	0.0052	0.5057
ResNet18	0.8462	0.7202	0.7337	0.7499	0.7034	0.0109	0.2966



Ablation Study

Model	AUROC	MaxF	AP	PRE	REC	FPR	FNR
UNet+a2	0.7843	0.6529	0.6941	0.7929	0.5753	0.0068	0.4247
PSPNet+a1	0.8476	0.7186	0.7337	0.7471	0.7064	0.0112	0.2936
DeepLabv3+a1	0.8589	0.7460	0.7584	0.7757	0.7281	0.0102	0.2719
Ours-rgb	0.6025	0.3083	0.5164	0.6901	0.2073	0.0024	0.7927
Ours-i	0.7687	0.5803	0.6127	0.6525	0.5510	0.0137	0.4490
Ours-rgbi	0.8744	0.7579	0.7675	0.7633	0.7601	0.0114	0.2399



Future Work

1. Generative & Foundation Models

- Diffusion: Conditional mask generation for wheel track recovery under severe noise.
- Autoencoder: Patch reconstruction to capture snow compression and track features.
- GAN: Feature contrast enhancement to extract low-contrast wheel track from snow cover.
- VLM: Zero-shot reasoning and open-vocabulary context for track identification.

2. Terrain & Weather Diversity

- Slush
- Black ice
- Muddy roads
- Flooded roads
- Rain
- Fog
- Dust